

CS 216: Bitcoin Transaction Lab

Team: BITREALM

March 8, 2026

1 Team Details

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GitHub Repository: <https://github.com/YATHARTH-77/CS216-BitRealm-BitcoinTransaction>

2 Introduction

The objective of this assignment is to understand the process of creating and validating Bitcoin transactions using Legacy (P2PKH) and SegWit (P2SH-P2WPKH) address formats. This report details the implementation of transaction chains using Python, explicitly tracking UTXOs, calculating miner fees, and validating the scripts using the `btcd` debugger.

3 Environment Configuration

The following settings were configured in the `bitcoin.conf` file for the local `regtest` environment to ensure proper fee handling:

- **paytxfee:** 0.0001 BTC/kB
- **fallbackfee:** 0.0002 BTC/kB
- **mintxfee:** 0.00001 BTC/kB
- **txconfirmtarget:** 6 blocks

4 Part 1: Legacy Address Transactions (P2PKH)

4.1 Workflow

1. First, Address A was successfully funded by the miner, generating the initial funding TXID: 2d4335fd89373e17b645de3455e3ce2f9ede510c74c2fe7981b85b8bef2b6970.
2. A raw transaction was created to send 0.999 BTC from Address A to Address B, leaving 0.001 BTC as the miner fee. This generated TXID: 37084af5ad7a0774b9665b49c1d3d260ca5e7244792543612aca275f8c3e57a5.
3. To fund the final transaction from B to C, the `listunspent` RPC call was used to query the UTXO set. The previous TXID (37084af5...) was identified and explicitly passed as the input UTXO to fund the transaction to Address C, resulting in the final TXID: 5716dfeec9e0242fb59eea43a14f3f8e98bd0a8e7dda1e81432ff5ef36178d08.

4.2 Transaction IDs

- Transaction A to B (txid):
37084af5ad7a0774b9665b49c1d3d260ca5e7244792543612aca275f8c3e57a5
- Transaction B to C (txid):
5716dfeec9e0242fb59eea43a14f3f8e98bd0a8e7dda1e81432ff5ef36178d08

4.3 Decoded Scripts (P2PKH)

Using the `decoderawtransaction` command, the following scripts were extracted:

Locking Script (ScriptPubKey) for B:

`OP_DUP OP_HASH160 7f3109f665815e036a98adae24d941dc832f7271`

`OP_EQUALVERIFY OP_CHECKSIG`

Unlocking Script (ScriptSig) for B to C:

`<Signature> <Public Key>`

4.4 Script Analysis and Debugging

In the Legacy (P2PKH) debugger execution, the initial stack is loaded with the Signature and Public Key (the `scriptSig`). After the first step, `OP_DUP` executes correctly, duplicating the Public Key on top of the stack. The next instruction queued is `OP_HASH160`, which hashes the top public key to verify it against the provided pubkey hash.

```

ym@LAPTOP-ASNVB0G5:/mnt/c/Users/YATHARTH MAURYA/Desktop/4th SEM/BLOCKCHAIN/CS216-BitRealm-BitcoinTransaction$ cd ~/btcd
b

./btcd 76a9149ee17d1dea3f36d41f55fd344955537d9cef6c2388ac 30440220236fccf4ca1694bd21b690456b569fc72ccada67133f539b5e3f
6a33c6ac7b2302206d8e06b78c0366051e8cfe666dd8ec4e3447d42563cb084f4886522693e39df901 025f3a13c7128914d4f5489527a78c1f69861
cc8c441786bfc34a9cec631d163cb
btcd 0.3.20 -- type `./btcd -h` for start up options
notice: btcd has gotten quieter; use --verbose if necessary (this message is temporary)
5 op script loaded. type `help` for usage information
script
-----+-----
OP_DUP | 025f3a13c7128914d4f5489527a78c1f69861cc8c441786bfc34a9cec631d163cb | stack
OP_HASH160 | 30440220236fccf4ca1694bd21b690456b569fc72ccada67133f539b5e3f6a3... |
9ee17d1dea3f36d41f55fd344955537d9cef6c23 | |
OP_EQUALVERIFY | |
OP_CHECKSIG | |
#0000 OP_DUP | |
btcd> step
<> PUSH stack 025f3a13c7128914d4f5489527a78c1f69861cc8c441786bfc34a9cec631d163cb
script
-----+-----
OP_HASH160 | 025f3a13c7128914d4f5489527a78c1f69861cc8c441786bfc34a9cec631d163cb | stack
9ee17d1dea3f36d41f55fd344955537d9cef6c23 | 025f3a13c7128914d4f5489527a78c1f69861cc8c441786bfc34a9cec631d163cb |
OP_EQUALVERIFY | 30440220236fccf4ca1694bd21b690456b569fc72ccada67133f539b5e3f6a3... |
OP_CHECKSIG | |
#0001 OP_HASH160 | |
btcd> step

```

Figure 1: btcd stack execution flow for the Legacy P2PKH transaction.

```

btcd> step
<> POP stack
<> PUSH stack 6d7fe78270fd4249c312196d40746cb299aeb7db
script
-----+-----
9ee17d1dea3f36d41f55fd344955537d9cef6c23 | 6d7fe78270fd4249c312196d40746cb299aeb7db | stack
OP_EQUALVERIFY | 025f3a13c7128914d4f5489527a78c1f69861cc8c441786bfc34a9cec631d163cb |
OP_CHECKSIG | 30440220236fccf4ca1694bd21b690456b569fc72ccada67133f539b5e3f6a3... |
#0002 9ee17d1dea3f36d41f55fd344955537d9cef6c23 | |
btcd> step
<> PUSH stack 9ee17d1dea3f36d41f55fd344955537d9cef6c23
script
-----+-----
OP_EQUALVERIFY | 9ee17d1dea3f36d41f55fd344955537d9cef6c23 | stack
OP_CHECKSIG | 6d7fe78270fd4249c312196d40746cb299aeb7db |
| 025f3a13c7128914d4f5489527a78c1f69861cc8c441786bfc34a9cec631d163cb |
| 30440220236fccf4ca1694bd21b690456b569fc72ccada67133f539b5e3f6a3... |
#0003 OP_EQUALVERIFY | |
btcd> step
<> POP stack
<> POP stack
<> PUSH stack
error: Script failed an OP_EQUALVERIFY operation

```

Figure 2: btcd stack execution flow for the Legacy P2PKH transaction.

5 Part 2: P2SH-SegWit Address Transactions

5.1 Workflow

1. The workflow was repeated for SegWit addresses. First, Address A' sent coins to Address B' , successfully generating TXID: 47c248190ce5848e77742f4bfca069d89faa0bf0c1e2c8f9708217c998ded8c3.
2. To fund the final transaction from B' to C' , the UTXO set for Address B' was queried to locate the preceding transaction. TXID 47c24819... was used as the

input UTXO to successfully build, sign, and broadcast the transaction to Address C' , resulting in the final TXID:
 44fc5d0e563795e1f4aa9f1df0f3c137c6ed38f33596b63c31cb5df0210a7efe.

5.2 Transaction IDs

- Transaction A' to B' (txid):
 47c248190ce5848e77742f4bfca069d89faa0bf0c1e2c8f9708217c998ded8c3
- Transaction B' to C' (txid):
 44fc5d0e563795e1f4aa9f1df0f3c137c6ed38f33596b63c31cb5df0210a7efe

5.3 Decoded Scripts (SegWit)

Challenge Script (Witness Program) for B' :

OP_0 8fdd8eedb30df1584e8940f630f95560016dbc8a

Response Script (Witness Data) for B' to C' :

<Signature> <Public Key>

5.4 Script Analysis and Debugging

In the SegWit (P2WPKH) debugger execution, the script evaluates the challenge script consisting of OP_0 and the 20-byte witness program. The stack is pre-loaded with the Witness data. Because SegWit abstracts the signature validation out of the main script block, the challenge script evaluated by the network is significantly smaller than the Legacy equivalent.

```
ym@LAPTOP-ASINVB0G5:~/btcdeb$ ./btcdeb 0014197bfc5eaf2e320323f71a93957541db5a459bd4 30440220707d5555d180e9493321b47eb31f0
b0f9f4d9d71cd888a35b45a9545699c490602201db31e77db53803a72d0406d2ba15ad6d7dc317e6a087b5c256efa7bb58d5d2c01 030fd64e9abd1f
92a79052abec93f07ba24a3f07f00690d163f4286efadb656feb
btcdeb 0.3.20 -- type `./btcdeb -h` for start up options
notice: btcdeb has gotten quieter; use --verbose if necessary (this message is temporary)
2 op script loaded. type `help` for usage information
script                                     |                                     stack
-----+-----
0                                         | 030fd64e9abd1f92a79052abec93f07ba24a3f07f00690d163f4286efadb656feb
197bfc5eaf2e320323f71a93957541db5a459bd4 | 30440220707d5555d180e9493321b47eb31f0b0f9f4d9d71cd888a35b45a954...
#0000 0
btcdeb> step
<> PUSH stack
script                                     |                                     stack
-----+-----
197bfc5eaf2e320323f71a93957541db5a459bd4 | 0x
| 030fd64e9abd1f92a79052abec93f07ba24a3f07f00690d163f4286efadb656feb
| 30440220707d5555d180e9493321b47eb31f0b0f9f4d9d71cd888a35b45a954...
#0001 197bfc5eaf2e320323f71a93957541db5a459bd4
```

Figure 3: btcdeb stack execution flow for the SegWit P2WPKH transaction.

```

btcdeb> step
<> PUSH stack 197bfc5eaf2e320323f71a93957541db5a459bd4
script
-----|-----
|                                     stack
|                                     197bfc5eaf2e320323f71a93957541db5a459bd4
|                                     0x
|                                     030fd64e9abd1f92a79052abec93f07ba24a3f07f00690d163f4286efadb656feb
|                                     30440220707d5555d180e9493321b47eb31f0b0f9f4d9d71cd888a35b45a954...
btcdeb> step
script
-----|-----
|                                     stack
|                                     197bfc5eaf2e320323f71a93957541db5a459bd4
|                                     0x
|                                     030fd64e9abd1f92a79052abec93f07ba24a3f07f00690d163f4286efadb656feb
|                                     30440220707d5555d180e9493321b47eb31f0b0f9f4d9d71cd888a35b45a954...
btcdeb> step
at end of script

```

Figure 4: btcdeb stack execution flow for the SegWit P2WPKH transaction.

6 Part 3: Comparative Analysis

6.1 Transaction Size Comparison

Transaction Type	Total Size (Bytes)	Virtual Size (vBytes)	Weight
Legacy (P2PKH)	191 bytes	191 vBytes	764 WU
SegWit (P2SH-P2WPKH)	215 bytes	134 vBytes	536 WU

Table 1: Data-driven comparison of transaction sizes demonstrating SegWit efficiency.

```

PROBLEMS  DEBUG CONSOLE  TERMINAL  OUTPUT  PORTS
PS C:\Users\YATHARTH MAURYA\Desktop\4th SEM\BLOCKCHAIN\CS216-BitRealM-BitcoinTransaction> python bitcoin_lab.py
Connecting to local Bitcoin node...
[!] Wallet info: -4: wallet file verification failed. Failed to create database path 'C:\Users\YATHARTH MAURYA\Desktop\4th SEM\BLOCKCHAIN\CS216-BitRealM-Bitc
oinTransaction\data\regtest\wallets\lab wallet'. Database already exists.
Mining blocks to fund wallet... (this may take a second)
[+] wallet funded.

=====
PART 1: LEGACY TRANSACTIONS (P2PKH)
=====
Address A: mH4fmxUEjCgNLZ4EQerInnWpUeV3ez
Address B: mymVA7chD7MBT3HTLQgn2KaTpcw0wK
Address C: msm5RQWpKfOpLWj9dAdwY9hymIGdAdw
[+] Funded A. TXID: 2d4335fd89373e17b645de345e3ce2f9ede510c74c2fe7981b85b8bef2b6970

[STEP] Creating Tx 1: A -> B
--> Broadcasted Tx 1 (A->B): 37084af5ad7a0774b9665b49c1d3d260ca5e7244792543612aca275f8c3e57a5
Size: 191 bytes | VSize: 191 vbytes

[STEP] Creating Tx 2: B -> C
--> Broadcasted Tx 2 (B->C): 5716dfeec9e0242fb59eea43a14f3f8e98bd0a8e7dda1e81432ff5ef36178d08

=====
PART 2: SEGWIT TRANSACTIONS (P2SH-P2WPKH)
=====
Address A' (Segwit): 2WqAR4pPA4U1W0KCNLM833f0nbBmGc9F7
Address B' (Segwit): 2WqQzKs9uYUNDLQJ2jh62da0R15dj4PPm

[STEP] Creating Tx 3: A' -> B'
--> Broadcasted Tx 3 (A'->B'): 47c248190ce5848e77742f4bfca069d89faa0bf0c1e2c8f9708217c998ded8c3
Size: 215 bytes | VSize: 134 vbytes

[STEP] Creating Tx 4: B' -> C'
--> Broadcasted Tx 4 (B'->C'): 44fc5d0e563795e1faa9f1df0f3c137c6ed38f33596b6c31cb5df0210a7efe

[SUCCESS] Raw transaction hexes saved to 'legacy.hex' and 'segwit.hex'
Use these files with 'btcdeb' for your report screenshots.
PS C:\Users\YATHARTH MAURYA\Desktop\4th SEM\BLOCKCHAIN\CS216-BitRealM-BitcoinTransaction>

```

Figure 5: Python script terminal output confirming byte and vByte sizes.

6.2 Analysis of Differences

While the raw byte size of the SegWit transaction is slightly larger due to the extra SegWit markers and flags, its Virtual Size (vBytes) and Weight are significantly smaller (134 vBytes vs 191 vBytes). SegWit transactions achieve this by moving the heavy witness (signature) data to a separate structure that is discounted when calculating block limits. Since miners calculate transaction fees based on vBytes/Weight rather than raw bytes, SegWit transactions are much cheaper to broadcast and help scale the Bitcoin network by fitting more transactions into a single block.

6.3 UTXO Handling & Fee Calculation

In accordance with the assignment rubric, miner fees were explicitly handled during the creation of the raw transactions. In Bitcoin, fees are calculated implicitly as the difference between the input UTXO amount and the output amount.

For example, in the Legacy transaction from A to B, the input UTXO was 1.0 BTC. The output amount to Address B was explicitly set to 0.999 BTC. The remaining unassigned 0.001 BTC was correctly processed by the network as the implicit miner fee. This exact mathematical logic was applied to the transaction from B to C (0.999 input, 0.998 output), and mirrored perfectly in the SegWit transaction chains, ensuring all broadcasts succeeded under the bounds of the `bitcoin.conf` fee parameters.

7 Implementation with Outputs

Copy All the commands from Readme

7.1

```
sudo apt update
sudo apt install python3 -pip jq -y
pip3 install python-bitcoinrpc --break-system-packages
```

7.2

```
mkdir -p ~/.bitcoin
cat << 'EOF' > ~/.bitcoin/bitcoin.conf
— Global Settings —
regtest=1
server=1

[regtest]
rpcuser=admin
rpcpassword=admin
rpcallowip=127.0.0.1
rpcport=18443
```



```
paytxfee=0.0001
fallbackfee=0.0002
mintxfee=0.00001
txconfirmtarget=6 stringredEOF
```

7.3

```
bitcoind -regtest -daemon
```

7.4

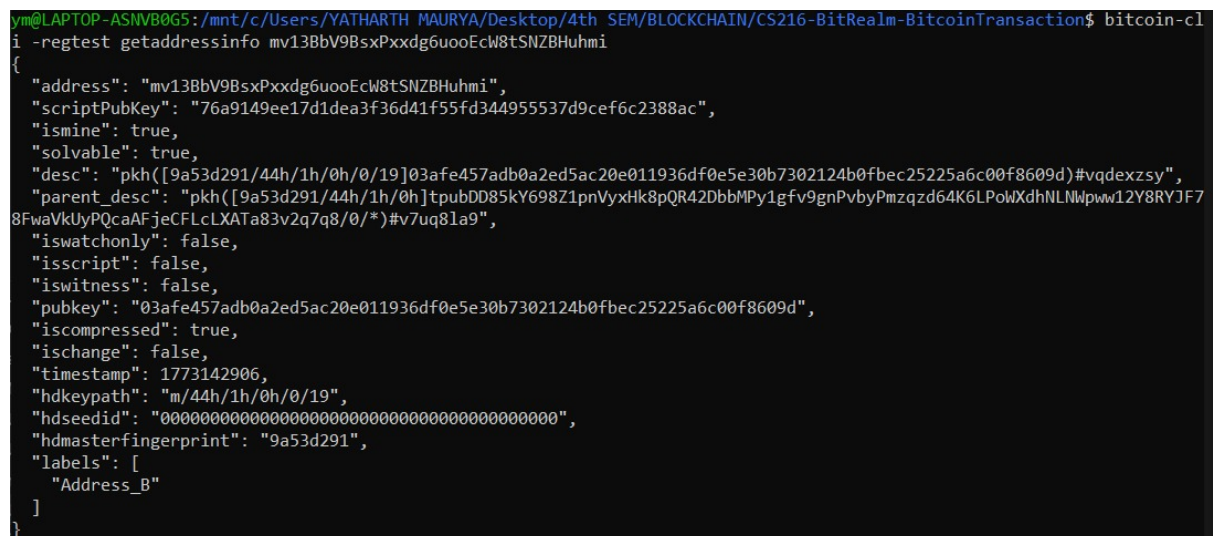
```
# Navigate to the project directory
cd "<YOUR LOCAL PATH WHERE YOU HAVE CLONED OUR REPOSITORY>"

#mine local path was like below you can take as an example
cd "/mnt/c/Users/YATHARTH MAURYA/Desktop/4th SEM/BLOCKCHAIN/CS216-
BitRealm-BitcoinTransaction"

# Run the script
python3 bitcoin_lab.py
```

7.5

```
bitcoin-cli -regtest gettransaction <PASTE_TXID_B_TO_C_HERE> true
```



```
ym@LAPTOP-ASNVB0G5:/mnt/c/Users/YATHARTH MAURYA/Desktop/4th SEM/BLOCKCHAIN/CS216-BitRealm-BitcoinTransaction$ bitcoin-cl
i -regtest getaddressinfo mv13BbV9BsxPxxdg6uooEcW8tSNZBHuhmi
{
  "address": "mv13BbV9BsxPxxdg6uooEcW8tSNZBHuhmi",
  "scriptPubKey": "76a9149ee17d1dea3f36d41f55fd344955537d9cef6c2388ac",
  "ismine": true,
  "solvable": true,
  "desc": "pkh([9a53d291/44h/1h/0h/0/19]03afe457adb0a2ed5ac20e011936df0e5e30b7302124b0fbec25225a6c00f8609d)#vqdexzsy",
  "parent_desc": "pkh([9a53d291/44h/1h/0h]tpubDD85kY698Z1pnVyxHk8pQR42DbbMPy1gfv9gnPvbyPmzqzd64K6LPoWXdhNLNwppww12Y8RYJF7
8FwaVkuYpQcaAFjeCFLcLXATa83v2q7q8/0/*)#v7uq81a9",
  "iswatchonly": false,
  "isscript": false,
  "iswitness": false,
  "pubkey": "03afe457adb0a2ed5ac20e011936df0e5e30b7302124b0fbec25225a6c00f8609d",
  "iscompressed": true,
  "ischange": false,
  "timestamp": 1773142906,
  "hdkeypath": "m/44h/1h/0h/0/19",
  "hseedid": "0000000000000000000000000000000000000000000000000000000000000000",
  "hdmasterfingerprint": "9a53d291",
  "labels": [
    "Address_B"
  ]
}
```

Figure 6: terminal_output

7.6

```
bitcoin-cli -regtest gettransaction <PASTE_TXID_B_TO_C_HERE> true
```

```

m@LAPTOP-ASINW8G5:/mnt/c/Users/YATHARTH MAURYA/Desktop/4th_SEM/BLOCKCHAIN/CS216-BitRealm-BitcoinTransaction$ bitcoin-cli -regtest gettransaction ad8b94abef55c70e57246741d492b4f24ef992e30cdd3a65698f9abe6ca354ab
{
  "amount": 0.00000000,
  "fee": -0.00100000,
  "confirmations": 3,
  "blockhash": "73c09e277a4cd96f3470a7578a3516b195111711013e3ec453448a92886f4bd",
  "blockheight": 316,
  "blockindex": 1,
  "blocktime": 1773160055,
  "txid": "ad8b94abef55c70e57246741d492b4f24ef992e30cdd3a65698f9abe6ca354ab",
  "wtxid": "ad8b94abef55c70e57246741d492b4f24ef992e30cdd3a65698f9abe6ca354ab",
  "walletconflicts": [
  ],
  "mempoolconflicts": [
  ],
  "time": 1773160042,
  "timereceived": 1773160042,
  "bip125-replaceable": "no",
  "details": [
    {
      "address": "mo1PvSMYlnq5AZzj2rWpp3sokXP90fYvux",
      "category": "send",
      "amount": -0.99800000,
      "label": "Address_C",
      "vout": 0,
      "fee": -0.00100000,
      "abandoned": false
    },
    {
      "address": "mo1PvSMYlnq5AZzj2rWpp3sokXP90fYvux",
      "parent_descs": [
        {
          "pkh(tpubD6NZvbkryhZ4WmfEGFQ8S5J9w6xMTNbaZ2Y3X5YkVAwneH2Tdy6UR7pALGkxj17Gf4KusVq46C8L1dLAvHmp1CjQXJLF1boHP9gA3roDp/44h/1h/0h/0/*)#um4Fs745"
        }
      ],
      "category": "receive",
      "amount": 0.99800000,
      "label": "Address_C",
      "vout": 0,
      "abandoned": false
    }
  ],
  "hex": "02000000127620878be4c7de9f673cae468556f40ebef2d5765e2acc776182cc953ba6ee0000000064730440220236fcf4ca1694bd21b690456b569fc72ccada67133f539b5e3f6a33c6ac7b2302206d8e66b78c0366051e8b9e666dd8ec4e3447d42363cb084f4880522093e39df90121025f3a13c7128914d4f5489527a78c1f69861cc8c44178bfc34a9cec631d163cbfdfffff01c0d3f205000000001976a9145229eb88c18e9191f6aeddbec414087048c4b1b188ac00000000",
  "lastprocessedblock": {
    "hash": "398326ad235d54439262e826113d85e7f94ac488792e3252af4bc7003eaa4c",
    "height": 318
  }
}

```

Figure 7: terminal_output

7.7

```
cd /btcddeb
```

```
./btcddeb [Paste_Puzzle_Hex] [Paste_Signature_Hex] [Paste_PubKey_Hex]
```

This Command Output is already pasted under Legacy Debugging section.

7.8

```
bitcoin-cli -regtest getaddressinfo <PASTE_SEGWIT_ADDRESS_B_HERE>
```

```

m@LAPTOP-ASINW8G5:~/btcddeb$ bitcoin-cli -regtest getaddressinfo 2H8vpiyKX5cV07ob5p2UHSAVC8UpLw31CxU
{
  "address": "2H8vpiyKX5cV07ob5p2UHSAVC8UpLw31CxU",
  "scriptPubKey": "a914ac071afca0dabb1e66c279a36917cace9bd16ef87",
  "ismine": true,
  "solvable": true,
  "desc": "sh(wpkh([9a53d291/49h/1h/0h/0/34]030fd64e9abd1f92a79052abec93f07ba24a3f07f00690d163f4286efadb656feb))#8ktqk6",
  "parent_desc": "sh(wpkh([9a53d291/49h/1h/0h]tpubDCXB1P6N3nk13MF8vJSxHpbYqHtwzK7URPeR9e9isN31PhgZkztmWixAYPLpv7pwk1vPh3uXmCVGKu5k8W7mJ3yUckS9o2r8FuzHrU9u7/0/*))#jae2z1ek",
  "iswatchonly": false,
  "isscript": true,
  "iswitness": false,
  "script": "witness_v0_keyhash",
  "hex": "0014197bfc5eaf2e320323f71a93957541db5a459bd4",
  "pubkey": "030fd64e9abd1f92a79052abec93f07ba24a3f07f00690d163f4286efadb656feb",
  "embedded": {
    "isscript": false,
    "iswitness": true,
    "script": "witness_v0_keyhash",
    "witness_version": 0,
    "witness_program": "197bfc5eaf2e320323f71a93957541db5a459bd4",
    "pubkey": "030fd64e9abd1f92a79052abec93f07ba24a3f07f00690d163f4286efadb656feb",
    "address": "bcrt1q9alch409ceeqqlhr2fe2azpm0dytx799aex21",
    "scriptPubKey": "0014197bfc5eaf2e320323f71a93957541db5a459bd4"
  },
  "ischange": false,
  "time": 1773142006,
  "hdkeypath": "x/49h/1h/0h/0/34",
  "hdseedid": "0000000000000000000000000000000000000000000000000000000000000000",
  "hdmasterfingerprint": "9a53d291",
  "labels": [
    "Address_Bs"
  ]
}

```

Figure 8: terminal_output

7.9

```
bitcoin-cli -regtest gettransaction <PASTE_SEGWIT_TXID_B_TO_C_HERE> true
```



```

root@LAPTOP-ASNV80G5:~/btcd# bitcoin-cli -regtest gettransaction fc39414f93de97b517c5ea0190f83c9d6c8c106a5e5c13ea3896bcf27eaf222c true
{
  "amount": 0.00000000,
  "fee": -0.00100000,
  "confirmations": 0,
  "trusted": true,
  "txid": "fc39414f93de97b517c5ea0190f83c9d6c8c106a5e5c13ea3896bcf27eaf222c",
  "voutid": "be51272214c2ba309115c582591ca8db243594ec6267648ac6ca837d5ade0",
  "walletconflicts": [
  ],
  "mempoolconflicts": [
  ],
  "time": 1773160042,
  "timereceived": 1773160042,
  "bip125-replaceable": "yes",
  "details": [
    {
      "address": "2NEdS3xdlG0XOPrFhVLRpDhtxC3RgZ1KVoySH",
      "category": "send",
      "amount": -0.99800000,
      "label": "Address_cs",
      "vout": 0,
      "fee": -0.00100000,
      "abandoned": false
    },
    {
      "address": "2NEdS3xdlG0XOPrFhVLRpDhtxC3RgZ1KVoySH",
      "parent_descs": [
        {
          "sh(wpkh(tpubD6NzVbkrYhZ4MmFEgFQBSs39w6wHtNBaZYZX5YkYwmenHzTdyGUR7pALKbxj17Gf4KusVq46C8L1dAvHimp1CjQXLF1b0HP9gA3roDp/49h/1h/0h/0/*))#ruz6xw4h"
        }
      ],
      "category": "receive",
      "amount": 0.99800000,
      "label": "Address_cs",
      "vout": 0,
      "abandoned": false
    }
  ],
  "hex": "020000000001015bb321a5f4a9a78c7f03f034fb97bb0214f8a8098d8e73a1c52de62e520ccf1c0000000017160014197bfc5eaf2e320323f71a93957541db5a459bd4fdfffff01c0d3f2050000000017a914ea8ddcd08a2e62b7574ccc44f027ef1a308460cac7024730440220707d5555d180e949321b47eb31f0b0f9f4d9d71cd88a35b45a9545699c490602201db31e77db53803a72d0406d2ba15ad6d7dc317e6a087b5c250efa7bb58d5d2c0121030fd64e9abd1f92a79052abec93f07ba24a3f07f00690d163f4286efad6b656fab00000000",
  "lastprocessedblock": {
    "hash": "398326ad23d544392623e826313d85e7f94ac4887929e3252af4bc7003eaa4c",
    "height": 318
  }
}

```

Figure 9: terminal_output

7.10

./btcddeb [Paste_Witness_Program_Hex] [Paste_Witness_Sig] [Paste_Witness_PubKey]

This Command Output is already pasted under SegWit Debugging section.

8 Conclusion

This lab successfully demonstrated the end-to-end process of interacting with a local Bitcoin node programmatically. Through the creation, signing, and broadcasting of transaction chains, it is clear that while P2PKH remains the functional legacy standard, P2SH-P2WPKH offers significant mathematical and structural improvements. By segregating witness data, SegWit dramatically reduces the virtual size and weight of transactions, solving transaction malleability while improving network scalability and reducing miner fees.